SHAZIN ANSARI

Associate AI Engineer | Generative AI, LLM Applications & RAG | Python, FastAPI, LangChain | Bengaluru, India

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Professional Summary

B.Tech (AI & Data Science) graduate and AI engineer with 1+ year of hands-on experience building LLM-powered and RAG-based applications in Python. Built RAG and conversational AI systems with LangChain, embeddings and vector search (FAISS, ChromaDB, pgvector), exposed through FastAPI and Flask REST APIs, containerized with Docker and backed by PostgreSQL and MySQL. Experienced in prompt engineering, LLM evaluation, hallucination reduction and tracing and debugging LLM pipelines with LangSmith, and in working with ML engineers and GenAI teams. Comfortable with Git, CI/CD concepts and AI-assisted development, and quick to learn new tools in a collaborative engineering team.

Technical Skills

- **Generative AI & LLM Applications:** LLM API integration (Groq API, Hugging Face, LLaMA), prompt engineering, conversational AI, chatbot and dialogue flow design, LLM evaluation, hallucination detection, faithfulness scoring
- **RAG & Agents:** RAG pipelines, document processing and chunking, embeddings, semantic search, vector databases (FAISS, ChromaDB, Pinecone, pgvector), LangChain, LangGraph, LlamaIndex, tool-based agent workflows
- **Backend & APIs:** Python, FastAPI, Flask, Django, building and consuming REST APIs, microservices, PostgreSQL, MySQL
- **DevOps & Monitoring:** Git, Docker, CI/CD integration, LangSmith tracing and LLM observability, debugging and latency benchmarking
- **Data & ML:** Pandas, NumPy, EDA, feature engineering, data processing, PyTorch, Scikit-learn, NLP
- **AI Development Tools:** Claude Code, Antigravity, Lovable, AI-assisted development

Projects

RAG-Based PDF Chatbot / *Python, FastAPI, LangChain, LangSmith, LLaMA, Groq API, FAISS, ChromaDB, pgvector, Docker*

- Reduced response latency by 30% and improved answer faithfulness, as measured by LangSmith tracing and hallucination evaluation, by building an end-to-end RAG application with document processing and chunking, embeddings, vector search and semantic retrieval, a FastAPI REST backend and a Dockerized deployment, and by using traces to find and fix pipeline issues.

Mental Health Chatbot / *Python, Flask, NLP, Semantic Search, REST API*

- Reached 85% query understanding accuracy across 50+ users, as measured by user testing, by designing conversational dialogue flows and building a Python backend with RESTful APIs for an NLP chatbot using semantic intent matching.

Multi-Disease Prediction System / *Django, Scikit-learn, PyTorch, Python*

- Achieved 80% accuracy, 80% precision and 88% recall on multi-class disease prediction, as measured by k-fold cross-validation, by building the model pipeline and deploying a Django backend with CI/CD integration.

Experience

Technical Research Associate – AI **Oct 2025 – Jun 2026**
Keywords Studios, Bengaluru

- Reduced hallucination rate and strengthened output reliability, as measured by faithfulness scores fed into iterative training cycles, by partnering with ML engineers on structured LLM evaluation and benchmarking of AI-generated outputs.
- Improved instruction-following and factual accuracy, as measured by structured ranking scores across response-quality dimensions, by contributing to RLHF-style alignment workflows.
- Accelerated model refinement decisions, as measured by the number of systematic failure modes surfaced to research teams, by identifying, documenting and escalating edge cases and errors in multimodal AI responses.
- Sustained consistent quality benchmarks across text and image modalities for multimodal foundation model training, as measured by team-wide output review coverage, by leading and coordinating task allocation for a team of 4 AI researchers.

AI Research Associate **Jul 2025 – Sep 2025**
Keywords Studios, Bengaluru

- Improved response quality and instruction adherence across state-of-the-art LLMs, as measured by consistency scores across diverse task types, by conducting structured prompt engineering and model evaluation.
- Shaped prompt design and model deployment decisions, as measured by evaluation insights adopted by cross-functional teams, by collaborating with GenAI teams on content generation and conversational workflows and explaining findings clearly in writing.
- Advanced RLHF and alignment research, as measured by volume of human preference feedback delivered, by evaluating LLM and multimodal AI outputs across text and image understanding tasks.

Data Science Intern **Jul 2024 – Apr 2025**
Future Optima IT Solutions, Kochi

- Achieved 85%+ classification accuracy on large datasets, as measured by model performance benchmarks, by processing data and building and optimizing decision tree and random forest models in Python with EDA and feature engineering.
- Maintained production model reliability, as measured by tracked accuracy and drift metrics, by collaborating with senior data scientists on algorithm implementation, monitoring and ongoing model evaluation.

Education

B.Tech – Artificial Intelligence & Data Science **2020 – 2024**
MES College of Engineering, Kuttipuram | APJ Abdul Kalam Technological University

Certifications

- Generative AI with LangChain & Hugging Face – Krish Naik, Udemy (2025)
- Career Essentials in Generative AI – Microsoft & LinkedIn (2024)